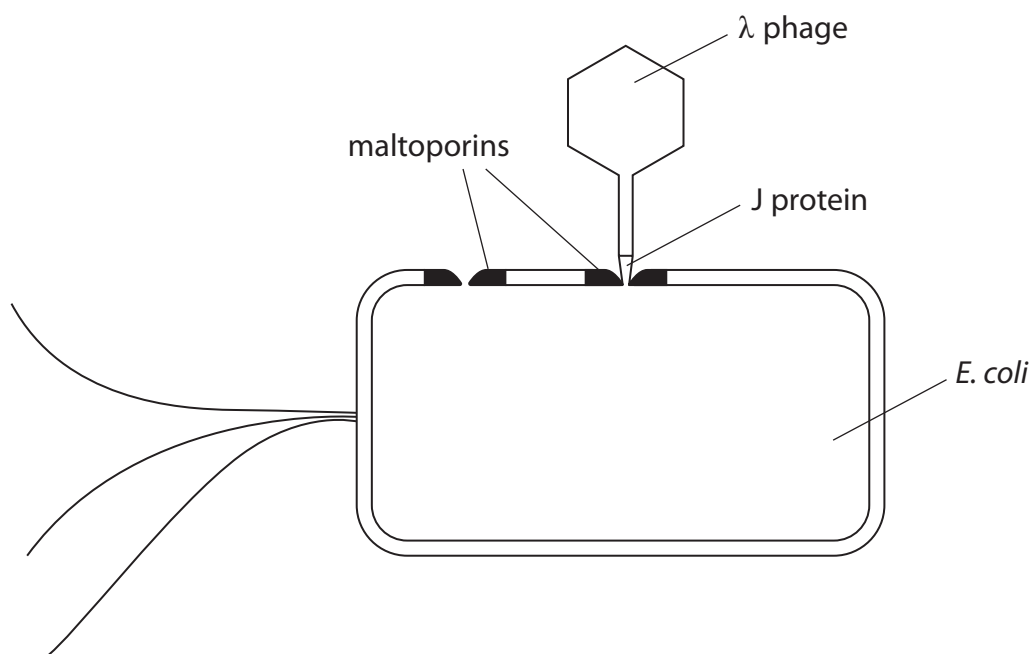


4 Pathogens continuously adapt to invade their host cells.

Host cells continuously adapt to avoid infection by pathogens.

This is an example of an 'evolutionary race'.

The diagram shows the pathogen lambda phage (λ phage) attached to its host cell, *E. coli*.



(a) How many of the following statements are correct for λ phage?

- the genetic material is DNA
- the capsid structure is described as complex
- the genetic material codes for the J protein

(1)

- ☐ **A** 0
- ☐ **B** 1
- ☐ **C** 2
- ☐ **D** 3

(b) Describe the role of the λ phage J protein.

Use the information in the diagram to support your answer.

(2)

(c) Maltoporin is coded for by the *lam β* gene of *E. coli*.

Maltoporin is involved in the transport of some sugars into *E. coli*.

(i) Give **one** reason why sugars are important to *E. coli*.

(1)

- *(ii) Discuss how *E. coli* and λ phage could interact and develop in their evolutionary race for existence.

Use all the information in this question to support your answer.

(6)

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(Total for Question 4 = 10 marks)